12.60

EE25BTECH11047 - RAVULA SHASHANK REDDY

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Question:

A square matrix $\mathbf{B}$ is skew-symmetric if

a) $\mathbf{B}^\top = -\mathbf{B}$ b) $\mathbf{B}^\top = \mathbf{B}$ c) $\mathbf{B}^{-1} = \mathbf{B}$ d) $\mathbf{B}^{-1} = \mathbf{B}^\top$

Solution:

$\mathbf{B}$ is a skew-symmetric, then by quadratic property

$$\mathbf{x}^\top \mathbf{B}\mathbf{x} = 0 \quad \forall \mathbf{x} \in \mathbb{R}^n \tag{1}$$

Consider two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$:

$$(\mathbf{x}+\mathbf{y})^\top \mathbf{B}(\mathbf{x}+\mathbf{y}) = \mathbf{x}^\top\mathbf{B}\mathbf{x} + \mathbf{x}^\top\mathbf{B}\mathbf{y} + \mathbf{y}^\top\mathbf{B}\mathbf{x} + \mathbf{y}^\top\mathbf{B}\mathbf{y} \tag{2}$$

$$\text{Since } \mathbf{x}^\top\mathbf{B}\mathbf{x} = \mathbf{y}^\top\mathbf{B}\mathbf{y} = 0, \text{ we have:} \tag{3}$$

$$\mathbf{x}^\top\mathbf{B}\mathbf{y} + \mathbf{y}^\top\mathbf{B}\mathbf{x} = 0 \quad \Rightarrow \quad \mathbf{x}^\top\mathbf{B}\mathbf{y} = -\,\mathbf{y}^\top\mathbf{B}\mathbf{x} \tag{4}$$

$$\text{Take } \mathbf{x} = \mathbf{e_1},\ \mathbf{y} = \mathbf{e_2} \text{ (standard basis vectors)} \tag{5}$$

$$\mathbf{e_1}^\top\mathbf{B}\mathbf{e_2} = -\,\mathbf{e_2}^\top\mathbf{B}\mathbf{e_1} \quad \Rightarrow \quad b_{ij} = -b_{ji} \tag{6}$$

$$\text{Since this holds for all } i, j, \text{ we conclude:} \tag{7}$$

$$\boxed{\mathbf{B}^\top = -\mathbf{B}} \tag{8}$$

Example:

$$\mathbf{B} = \begin{pmatrix} 0 & 2 & -1 \\ -2 & 0 & 3 \\ 1 & -3 & 0 \end{pmatrix}$$

$$\mathbf{B}^\top = \begin{pmatrix} 0 & -2 & 1 \\ 2 & 0 & -3 \\ -1 & 3 & 0 \end{pmatrix} = -\,\mathbf{B}$$